

EECS 496

Intellectual Property

November 1, 2024



WHAT IS IP?

Any creation of the mind.



WHAT IS IP?

ideas

art

literary works

inventions

branding

designs

music



IP Rights

Government-granted property rights in ideas and creative expressions

- Trademarks
- Copyrights
- Trade secrets
- Patents

Origin of US Patent Rights and Copyrights

US Constitution, Article 1, Section 8, Clause 8

“Congress shall have the power ... **to promote the progress of science and useful arts**, by securing for limited times to authors and inventors the exclusive right to their respective writings and discoveries.”



IP Rights – Why It's Relevant to You

- Senior design projects
- Employment agreements
- Startups

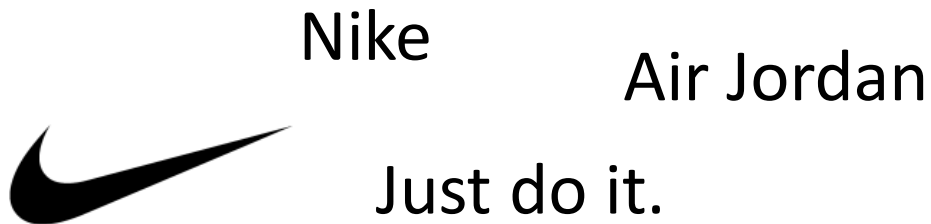
IP Rights

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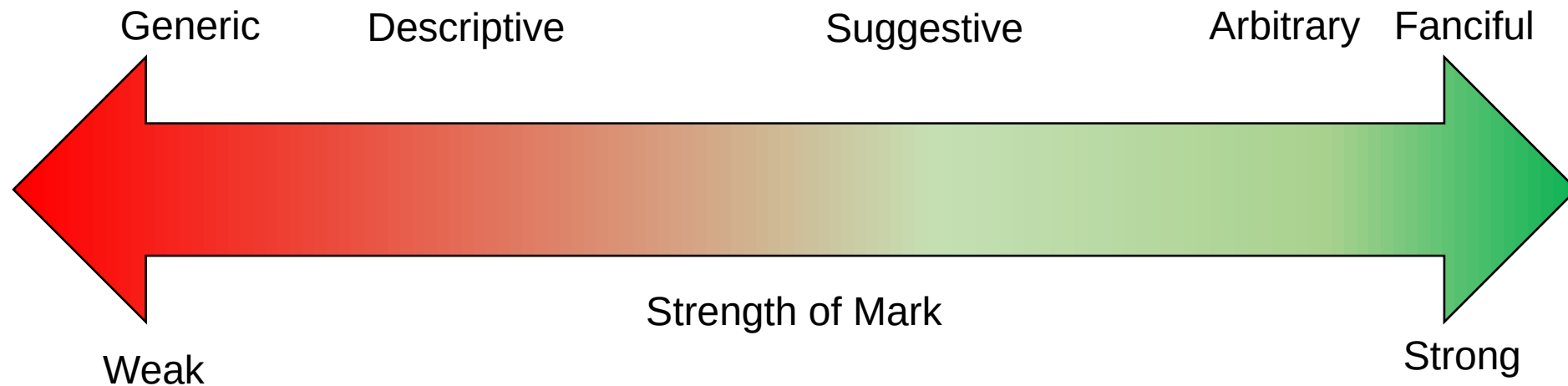
Trademarks

- Protects: a unique identifier of goods and services
- US trademark identified by: TM or SM or ®
- Protects against: use of confusingly similar marks in commerce
- Duration: indefinite (if in commercial use)



Trademarks

- US Requirements:
 - Unique identifier of goods or services
 - Used in interstate commerce



Trademarks

Guess which mark is NOT protected by a federal trademark

A “A sweet, slightly musky, vanilla **fragrance**, with slight overtones of cherry, combined with the smell of a salted, wheat-based dough”



C “A sound mark comprising a musical composition featuring two sixteenth note timpani strikes on D2 and D3, simultaneously with which are played three dotted half notes on D2, D4, and D5”

D “Drive Safely” slogan by Volvo

Copyrights

- Protects: original works of creative expression
 - Literary works, music, movies, paintings, photos, websites, software code, etc.
- Identified by: ©
- Protects against: copying
- Duration:
 - For individuals: life of creator + 70 years
 - For businesses: Shorter of...
 - 95 years from publication, or
 - 120 years from creation



Copyrights

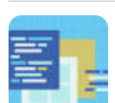
- US requirements:
 - Original work of authorship with modicum of creativity
 - Fixed in a tangible medium
- No registration required
- Advantages of registering:
 - Can enforce in court
 - Statutory damages
- Owner: author
 - Unless work-made-for-hire or assigned by contract

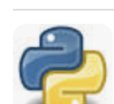


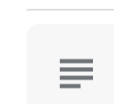
Copyrights & Open Source Licenses

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Trade Secrets

- Protects: proprietary information that derives economic value by being kept secret
 - formulations, processes, algorithms, business plans, project plans, customer lists, data, etc.
- Identified by: N/A
- Protects against: misappropriation of the secret (i.e., acquiring through a wrongful or nefarious act)
- Duration: as long as reasonable measures are taken to protect the information



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Trade Secrets

- US requirements:
 - (1) the information must have economic value;
 - (2) because it is not generally known; and
 - (3) the owner has taken reasonable measures to keep it secret
- Reasonable measures:
 - Have NDAs signed;
 - Mark trade secrets;
 - Limit employee access;
 - Cut off departing employees' access;
 - Use protected IT systems, passwords, policies for locking computers, physical security, etc.



Patents

Design Patents

Plant Patents

United States Patent [19] **Patent Number:** **Plant 7,197**
Luby et al. [45] **Date of Patent:** **Mar. 20, 1990**

[54] **APPLE TREE: HONEYCRISP**
 [75] Inventors: **Jim Luby, St. Paul; David S. Bedford, Delano, both of Minn.**
 [73] Assignee: **Regents of the University of Minnesota, Minneapolis, Minn.**
 [21] Appl. No.: **268,363**
 [22] Filed: **Nov. 7, 1988**
 [51] Int. Cl.⁴ **A01H 5/00**
 [52] U.S. Cl. **Plt./34**
 [58] Field of Search **Plt./34**

[56] **References Cited**
DUBLICATIONS

Extension Service Publication AG-FO-1111, 1987, pp. 1-3.

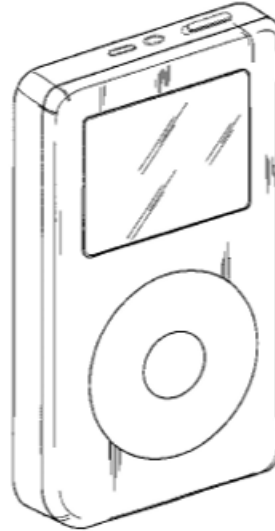
Primary Examiner—James R. Feyrer
Attorney, Agent, or Firm—Merchant, Gould, Smith, Edell, Welter & Schmidt

[57] ABSTRACT

An apple tree having a slightly upright form with moderately vigorous growth; a superior fruit with solid to mottled red color over a yellow background and a crisp and juicy fruit texture; an extended season of harvest but with generally even ripening of fruit so that repeat pickings are unnecessary; good hardiness with little winter injury when grown near Excelsior, Minn. at approximately 45° latitude; and having annual fruit production with no flower thinning required.



(10) **Patent No.:** **US D538,822 S**
 (45) **Date of Patent:** **** *Mar. 20, 2007**



Utility Patents

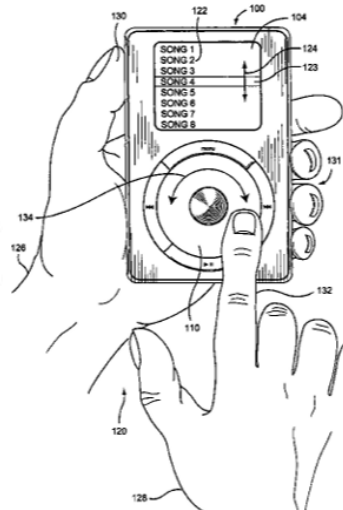
(12) **United States Patent**
Zadesky et al.

(54) **TOUCH PAD FOR HANDHELD DEVICE**

(75) Inventors: **Stephen Paul Zadesky, Portola Valley, CA (US); Tang Yew Tan, San Francisco, CA (US)**

(10) **Patent No.:** **US 7,333,092 B2**
 (45) **Date of Patent:** ***Feb. 19, 2008**

A media device for storing and playing media such as audio, video or images, includes a memory device configured to store a plurality of media items in a digital format. The media device also includes a display configured to present a group of media items from the plurality of stored media items and to present a visual indicator that is capable of scrolling through the displayed group of media items in order to designate a specific media item from the group of media items. The media device further includes a touch pad configured to receive input from a sliding motion or a tapping motion of a finger. The sliding motion of the finger controls the movement of the visual indicator through the group of media items. The tapping motion of the finger selects the specific media item that is designated by the visual indicator.



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Patents - Design

- Protects: inventive ornamental features
- Identified by: patent number (e.g., USD 670,090)
- Protects against: making, using, selling, or importing claimed invention
- Duration: 15 years from date of grant

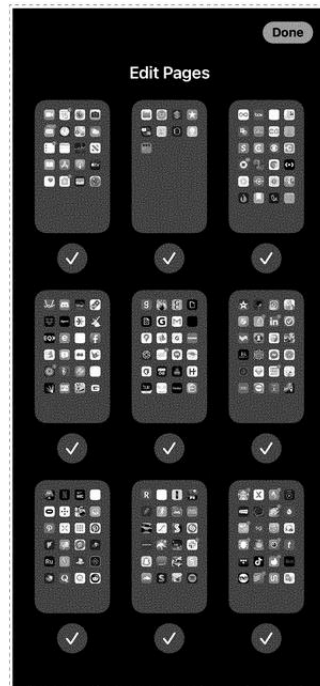
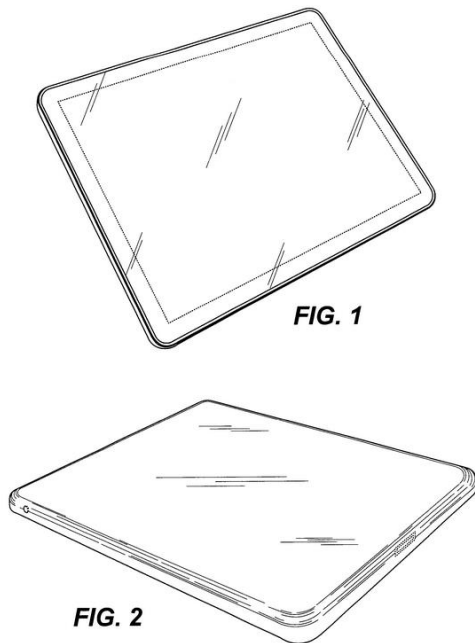
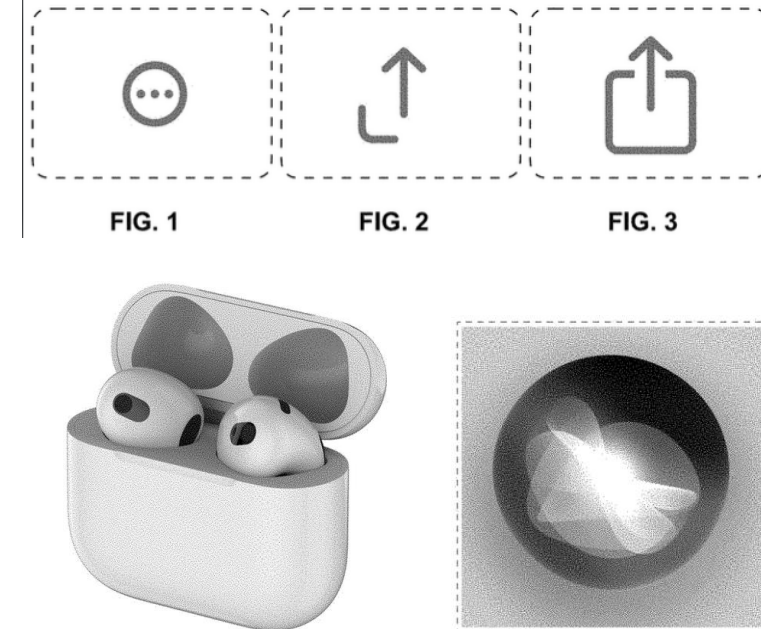
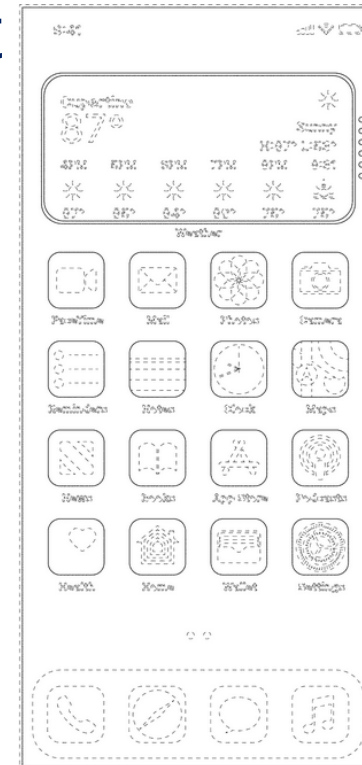


FIG. 1



FIG. 1



Patents - Utility

- Protects: technological inventions
 - Functional features
- Identified by: patent number (e.g., US Pat. No. 11,700,903)
- Protects against: making, using, selling, or importing claimed invention
- Duration: 20 years from filing

Patents - Utility

- US requirements:
 - Must file a patent application & pay \$\$\$
 - Evaluated to ensure it is:
 - Patentable eligible subject matter
 - Useful
 - Novel
 - Non-obvious
 - Owner: inventor
 - Unless assigned by contract

What is a utility patent? Specification

Claims

Title page

United States Patent

Yuen et al.

Patent No.:

US 8,180,591 B2

Date of Patent:

May 15, 2012

(54)

PORTABLE MONITORING DEVICES AND METHODS OF OPERATING SAME

(75)

Inventors: Shelten Gee Jao Yuen, Berkeley, CA (US); James Park, Berkeley, CA (US); Eric Nathan Friedman, San Francisco, CA (US)

(73)

Assignee: Fitbit, Inc., San Francisco, CA (US)

(*)

Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21)

Appl. No.: 13/246,843

(22)

Filed: Sep. 27, 2011

(65)

Prior Publication Data

US 2012/0084053 A1 Apr. 5, 2012

Related U.S. Application Data

(62) Division of application No. 13/156,304, filed on Jun. 8, 2011.

(60) Provisional application No. 61/388,595, filed on Sep. 30, 2010, provisional application No. 61/390,811, filed on Oct. 7, 2010.

(51)

Int. Cl. G06F 19/00 (2011.01)

(52)

U.S. Cl. 702/160

(58)

Field of Classification Search 702/150, 702/152, 160, 182-185, 188

See application file for complete search history.

(56)

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4,312,358 A 1/1982 Barney

4,367,752 A 1/1983 Jimenez et al.

4,578,769 A 3/1986 Frederick

4,977,509 A 12/1990 Pitchford et al.

5,058,427 A 10/1991 Brandt

26 Claims, 50 Drawing Sheets

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PORTABLE MONITORING DEVICES AND METHODS OF OPERATING SAME

RELATED APPLICATION

This application is a divisional of U.S. patent application Ser. No. 13/156,304, filed on Jun. 8, 2011, entitled "Portable Monitoring Devices and Methods of Operating Same" (still pending). This non-provisional application claims priority to U.S. Provisional Application No. 61/388,595, entitled "Portable Monitoring Devices and Methods of Operating Same", filed Sep. 30, 2010, and U.S. Provisional Application No. 61/390,811, entitled "Portable Monitoring Devices and Methods of Operating Same", filed Oct. 7, 2010; the contents of these U.S. Provisional Applications are incorporated by reference herein in their entirety.

FOREIGN PATENT DOCUMENTS

JP 11347021 12/1999

OTHER PUBLICATIONS

"Automatic classification of ambulatory movements and evaluation of energy consumptions utilizing accelerometers and a barometer", Ohtaki, et al, Microsystem Technologies, vol. 11, No. 8-10, Aug. 2005, pp. 1034-1040.

"Classification of Human Moving Patterns Using Air Pressure and Acceleration", Sagawa, et al, Proceedings of the 24th Annual Conference of the IEEE Industrial Electronics Society, vol. 2, Aug.-Sep. 1998, pp. 1214-1219.

"Non-restricted measurement of walking distance", Sagawa, et al, IEEE Int'l Conf. on Systems, Man, and Cybernetics, vol. 3, Oct. 2000, pp. 1847-1852.

(Continued)

Primary Examiner — Edward Raymond

(74) Attorney, Agent, or Firm — Neil A. Steinberg

(57)

ABSTRACT

The present inventions, in one aspect, are directed to a portable monitoring device comprising (i) a housing having a physical size and shape that is adapted to couple to the user, (ii) a motion sensor, disposed in the housing, to detect motion of the user and, in response thereto, to generate motion data, (iii) an altitude sensor, disposed in the housing, to sample an altitude of the user in response to a sample signal, wherein, in response to sampling the altitude of the user, the altitude sensor generates change in altitude data, and (iv) processing circuitry, disposed in the housing, to: generate the sample signal using the motion data, and calculate a number of stairs or flights of stairs traversed by the user using the motion data and the change in altitude data.

The present inventions relate to portable monitoring devices, and method of operating and controlling same, wherein the portable monitoring devices include an altitude sensor, motion sensor and processing circuitry to calculate, assess and/or determine calorie burn and/or other activity-related quantities of the user (for example, a human or non-human animal such as a dog, cat or horse). In these aspects, the present inventions employ the altitude sensor data and the motion sensor data to calculate, assess and/or determine the calorie burn and/or other activity-related quantities of the user (for example, number of steps and/or stairs, number of stair flights, elevation gain/loss from ambulatory and/or non-ambulatory locomotion, absolute elevation, elevation and/or activity points, activity intensity, distance traveled and/or pace, number of swim strokes and/or kicks, strokes per lap, lap time, pace and/or distance, number of pedal rotations of a bicycle, arm or wheel rotation of a wheelchair, heart rate, heart rate variability, respiration rate, stress levels, skin temperature, body temperature). In the following disclosure, use of the term "activity" includes sedentary and nonsedentary activities. As such, the present inventions also may be used to monitor activities related to sleeping, lying, sitting, and standing stationary and may provide corresponding metrics (for example, time asleep, the onset, duration, and number of awakenings while attempting to sleep, the time spent in various stages of sleep, sleep latency, sleep efficiency and other sleep quality parameters, the presence of sleep apnea and other diagnostic measures, time spent in a prone non-standing state, and resting heart rate).

In other aspects, the portable monitoring device of the present inventions may include a physiological sensor, in addition to the altitude sensor, motion sensor and processing circuitry. In these aspects, the present inventions employ the physiological sensor data, altitude sensor data and motion sensor data to calculate, assess and/or determine the calorie burn and/or other activity-related quantities of the user.

In certain aspects the processing circuitry is partially or wholly disposed external to the portable monitoring device wherein the external processing circuitry receives partially processed or "raw" sensor data. Here, the external processing circuitry partially or wholly calculates, assesses and/or determines the calorie burn and/or other activity-related quantities of the user.

Notably, the present inventions also relate to techniques or methods of calculating, assessing and/or determining the calorie burn and/or other activity-related quantities of the user

2

based on or using sensor data acquired by a portable monitoring device, for example, devices according to any of the of the present inventions.

SUMMARY

There are many inventions described and illustrated herein. The present inventions are neither limited to any single aspect nor embodiment thereof, nor to any combinations and/or permutations of such aspects and/or embodiments. Moreover, each of the aspects of the present inventions, and/or embodiments thereof, may be employed alone or in combination with one or more of the other aspects of the present inventions and/or embodiments thereof. For the sake of brevity, many of those permutations and combinations will not be discussed separately herein.

Importantly, this Summary may not be reflective of or correlate to the inventions protected by the claims in this or continuation/divisional applications hereof. Even where this Summary is reflective of or correlates to the inventions protected by the claims hereof, this Summary may not be exhaustive of the scope of the present inventions.

In a first aspect, the present inventions are directed to a portable monitoring device to calculate calorie burn or expenditure of a user wherein the portable monitoring device includes a housing having a physical size and shape that is adapted to couple to the body of the user. In this first aspect, the portable monitoring device includes a motion sensor to detect motion of the user and, in response, to generate data which is representative of motion of the user, and an altitude sensor to detect a change in altitude of the user and, in response, to generate data which is representative of the change in altitude of the user. The portable monitoring device may also include processing circuitry, coupled to the motion sensor and the altitude sensor, to (a) calculate data which is representative of a change in elevation of the user using (i) the data which is representative of motion of the user and (ii) the data which is representative of a change in altitude of the user, (b) calculate first calorie burn using the data which is representative of motion of the user, and (c) calculate second calorie burn by adjusting the first calorie burn based on data which is representative of a change in elevation of the user.

The portable monitoring device may also include a display, coupled to the processing circuitry, to output the second calorie burn.

In one embodiment, the processing circuitry of the portable monitoring device determines a change in elevation of the user when (i) the data which is representative of motion of the user exceeds a first threshold and (ii) the data which is representative of a change in altitude of the user exceeds a second threshold.

The processing circuitry may also calculate data which is representative of an accumulation of changes in elevation of the user by summing data which is representative of changes in elevation of the user in response to detecting a predetermined motion of the user based on the data which is representative of motion of the user. Here, the processing circuitry may calculate an altitude metric (for example, a number of stairs and/or a number of flights of stairs) using the data which is representative of the accumulation of changes in elevation of the user. A display, may output the altitude metric.

In another embodiment, the processing circuitry may also calculate data which is representative of an increase and/or decrease in elevation for a given time period using the data which is representative of a change in elevation of the user, and a display may output the data which is representative of

45

tion or image may thereafter be used in device fabrication, for example, by enabling generation of one or more masks that are used to form various components of the circuits in a device fabrication process.

Moreover, the various circuits and circuitry, as well as techniques, disclosed herein may be represented via simulations and simulation instruction-based expressions using computer aided design, simulation and/or testing tools. The simulation of the various sensors, processing circuitry, user interface, transmitter circuitry and/or receiver circuitry of the present inventions (regardless of combination or permutation of sensors, processing circuitry, transmitter circuitry and/or receiver circuitry), including the processes or techniques implemented thereby, may be implemented by a computer system wherein characteristics and operations of such circuitry, and techniques implemented thereby, are simulated, imitated, replicated, analyzed and/or predicted via a computer system. The present inventions are also directed to such simulations and testing of the inventive portable monitoring device (or portions thereof including, for example, the various sensors, processing circuitry, user interface, input/output circuitry (although not illustrated—the input/output circuitry may be discrete circuitry or circuitry which is integrated into the processing circuitry), transmitter circuitry and/or receiver circuitry), and/or techniques implemented thereby, and, as such, are intended to fall within the scope of the present inventions. The computer-readable media and data corresponding to such simulations and/or testing tools are also intended to fall within the scope of the present inventions.

The term "calculate" and other forms (i.e., calculating, calculated and calculation) in the claims means, among other things, calculate, assesses, determine and/or estimate and other forms thereof. In addition, the term "calorie burn" in the claims means, among other things, calorie burn or calorie expenditure and/or energy burn or energy expenditure—or the like.

Further, in the claims, the phrase "data which is representative of a change in altitude" means data which is representative of an altitude of the user (absolute altitude) and data which is representative of a change in altitude (relative altitude). Further, in the claims, the phrase "a change in altitude" means a change in altitude or height. Moreover, for the avoidance of doubt, in the claims, the term "flights of stairs" means "flights of stairs", "floors" and the like.

Notably, the terms "first", "second," and the like, herein do not denote any order, quantity, or importance, but rather are used to distinguish one element from another. Moreover, in the claims, the terms "a" and "an" herein do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item.

What is claimed is:

1. A portable monitoring device, adapted to couple to a body of a user, to calculate a number of stairs or flights of stairs traversed by the user, the portable monitoring device comprising:

a housing having a physical size and shape that is adapted to couple to the body of the user;

a motion sensor, disposed in the housing, to detect motion of the user and, in response thereto, to generate data which is representative of motion of the user;

an altitude sensor, disposed in the housing, to sample an altitude of the user in response to a sample signal, wherein, in response to sampling the altitude of the user, the altitude sensor generates data which is representative of a change in altitude of the user; and

processing circuitry, disposed in the housing and coupled to the motion sensor and the altitude sensor, to:

46

generate the sample signal using the data which is representative of motion of the user, and calculate a number of stairs or flights of stairs traversed by the user using the data which is representative of motion of the user and the data which is representative of a change in altitude of the user.

2. The portable monitoring device of claim 1 wherein the processing circuitry:

calculates a step frequency of the user using the data which is representative of motion of the user, and generates the sample signal based on the step frequency of the user.

3. The portable monitoring device of claim 1 wherein the processing circuitry detects a motion event of the user based on data which is representative of motion of the user and, in response, generates the sample signal.

4. The portable monitoring device of claim 1 wherein the processing circuitry generates the sample signal in response to determining a step by the user and/or after a predetermined period of time.

5. The portable monitoring device of claim 1 wherein the processing circuitry calculates the number of stairs or flights of stairs traversed by the user using (i) data which is representative of a change in altitude per step and (i) data which is representative of a change in altitude for a given time period.

6. The portable monitoring device of claim 1 wherein the processing circuitry determines the user traverses one or more stairs by calculating a change in altitude over one or more steps of the user using (i) data which is representative of motion of the user and (ii) data which is representative of a change in altitude of the user.

7. The portable monitoring device of claim 1 wherein the processing circuitry determines the user traverses one or more stairs based on (i) a change in altitude of the user per step exceeding a first threshold and (ii) a change in altitude of the user over a given time period exceeding a second threshold.

8. The portable monitoring device of claim 1 wherein the processing circuitry determines the user traverses one or more stairs based on (i) a change in altitude of the user per step being within a first threshold range and (ii) a change in altitude of the user over a given time period being within a second threshold range.

9. The portable monitoring device of claim 1 wherein the processing circuitry determines the user traverses one or more stairs based on evaluating (i) a change in altitude in relation to a first threshold, and (ii) a change in altitude of the user per step in relation to a second threshold range and (iii) a change in altitude of the user over a given time period in relation to a third threshold range.

10. The portable monitoring device of claim 1 wherein the processing circuitry:

determines a change in elevation of the user when (i) the data which is representative of motion of the user exceeds a motion-related threshold and (ii) the data which is representative of a change in altitude of the user exceeds an altitude-related threshold, and calculates data which is representative of an accumulation of changes in elevation of the user.

11. A portable monitoring device, adapted to couple to a body of a user, to calculate a number of stairs or flights of stairs traversed by the user, the portable monitoring device comprising:

a housing having a physical size and shape that is adapted to couple to the body of the user;

a motion sensor, disposed in the housing, to sample motion of the user in response to a motion sample signal,

Figures

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Patent Claims

- A series of sentences found at the end of the patent
- Define the metes and bounds of the protected invention
 - Define the limits of exactly what is covered by the patent
- Types of claims:
 - Independent claims
 - stand alone; independently enforceable
 - Dependent claims
 - only have meaning and force when combined with claims they depend from

What is patentable?

Requirements:

- 1) Patent Eligible Subject Matter
- 2) Useful
- 3) Enablement/Written description
- 4) Novel
- 5) Nonobvious

What is patent eligible subject matter?

35 U.S. Code § 101 - Inventions patentable

Whoever invents or discovers any new and useful **process, machine, manufacture, or composition of matter**, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

Difficulties with §101 and Software

- Ineligible: Algorithms, abstract ideas, naturally occurring phenomena
 - No accepted definition of “abstract idea”
 - Business methods are often held to be abstract ideas
- For software to be patent eligible, it must do “something more” than merely perform a business method
- A concrete, **specific** application of an abstract idea may be “something more”
 - Tied/performed to a specialized device
 - A **specific** element that addresses a **specific** problem by performing a **specific** function in a **specific** way to achieve a **specific** result
 - A technical problem to a technical solution

What is enablement/written description?

35 U.S. Code § 112(a)

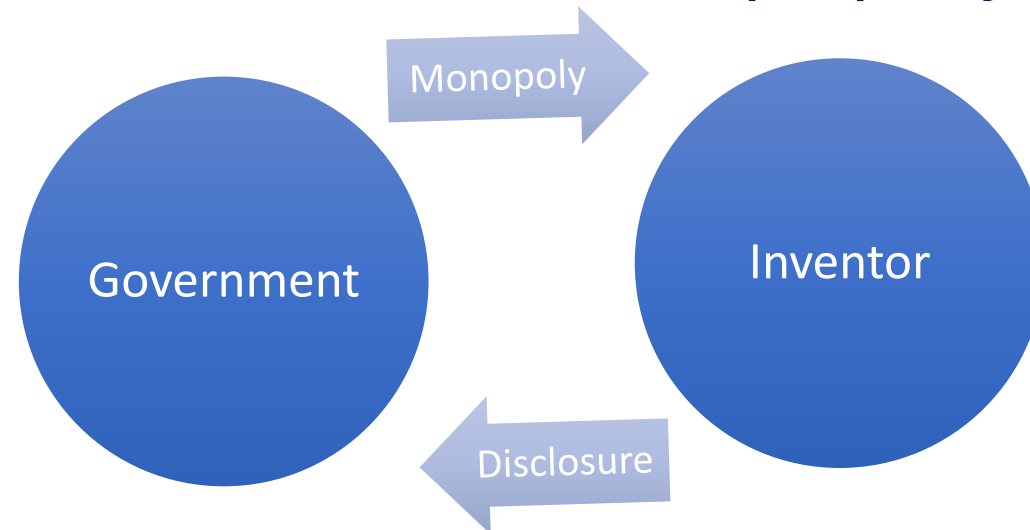
The specification shall contain a written description of the invention, and of the manner and process of making and using it, **in such full, clear, concise, and exact terms as to enable any person skilled in the art** to which it pertains, or with which it is most nearly connected, **to make and use the same**, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

What is Patentability?

- **Novel:** New; unique; never been disclosed by others before
 - Does a single reference disclose every feature of your “invention”?
- **Non-obvious:** “sufficiently inventive”
 - Does a combination of references disclose all features? If so, is there a motivation to combine those references?
 - Is the invention created by combining known elements according to known methods or by substituting one known element for another, and did it achieve predictable results?
 - Is a known technique used to improve a similar device in the same way?
 - Did the invention result from selecting a solution from a finite number of identifiable, predictable solutions having a reasonable expectation of success?

Why get a patent?

- Creates a limited monopoly around your product
 - Important to investors – provides some reassurance that they will see a return on their investment
- Marketing tool
- Can be sold and licensed like other property



Rights Granted Under Patent Law

- A patent grants its owner the right to EXCLUDE others from:
 - Making, Using, Selling, Importing, or Offering for sale the CLAIMED invention (during life of the patent in the country of patent issuance)
- It does NOT grant the RIGHT to make, use, sell, import, or offer for sale
 - Freedom to Operate is NOT guaranteed!

Patentability vs. Freedom to Operate

Patentability vs. Freedom to Operate

- Patentability:
 - Can you get a patent on your technology to exclude others?
 - Is your technology new and non-obvious?
 - Has anyone anywhere publicly disclosed the technology before?
- Freedom to Operate:
 - Can you practice your technology without infringing existing patents?
 - Are portions of your technology already patented?
 - Does your product include any technology that is described/recited in a claim of an active patent in a country where you plan to make/use/sell?

Patentability vs. Freedom to Operate USB Light Example

United States Patent [19]
Knoll et al.

[11] **Patent Number:** 5,623,610
[45] **Date of Patent:** Apr. 22, 1997

[54] **SYSTEM FOR ASSIGNING GEOGRAPHICAL ADDRESSES IN A HIERARCHICAL SERIAL BUS BY ENABLING UPSTREAM PORT AND SELECTIVELY ENABLING DISABLED PORTS AT POWER ON/RESET**

[75] **Inventors:** Shaun Knoll, Portland, Oreg.; Jeff C. Morriss, Boulder Creek, Calif.; Shelagh Callahan, Beaverton, Oreg.; Ajay V. Bhatt, El Dorado Hills, Calif.; Sudarshan B. Cadambi, Portland, Oreg.

[73] **Assignee:** Intel Corporation, Santa Clara, Calif.

[21] **Appl. No.:** 332,375

[22] **Filed:** Oct. 31, 1994

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5,130,983	7/1992	Heffner, III	370/85.8
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5,179,670	1/1993	Farmwald et al.	395/325
5,237,690	8/1993	Bealkowski et al.	395/700
5,257,160	10/1993	Yokohama et al.	361/246
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5,282,202	1/1994	Bernstein et al.	370/94.1
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Dean McCarron, Plug-and Play Specification Introduced: Microsoft, Intel, Others Announce User-Friendly ISA BUS, 7 Microprocessor Report 224 May 1993.

(List continued on next page.)

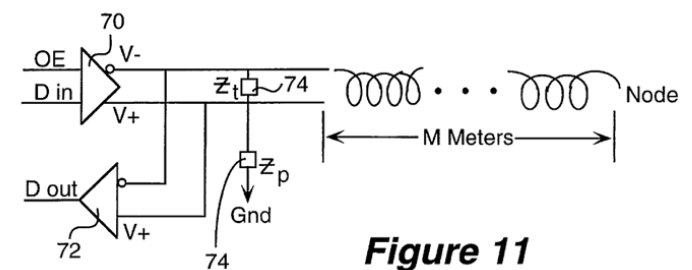
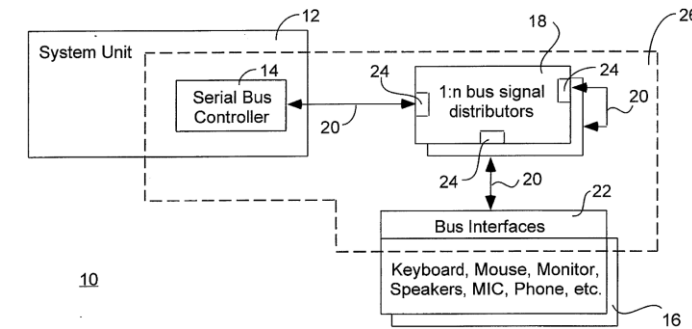
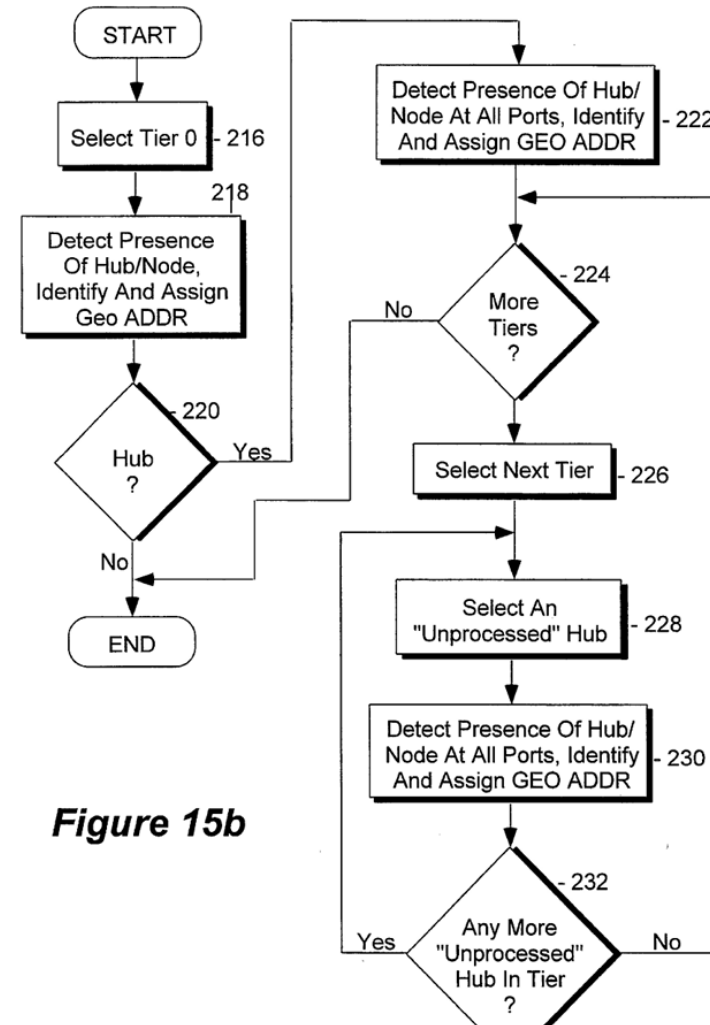
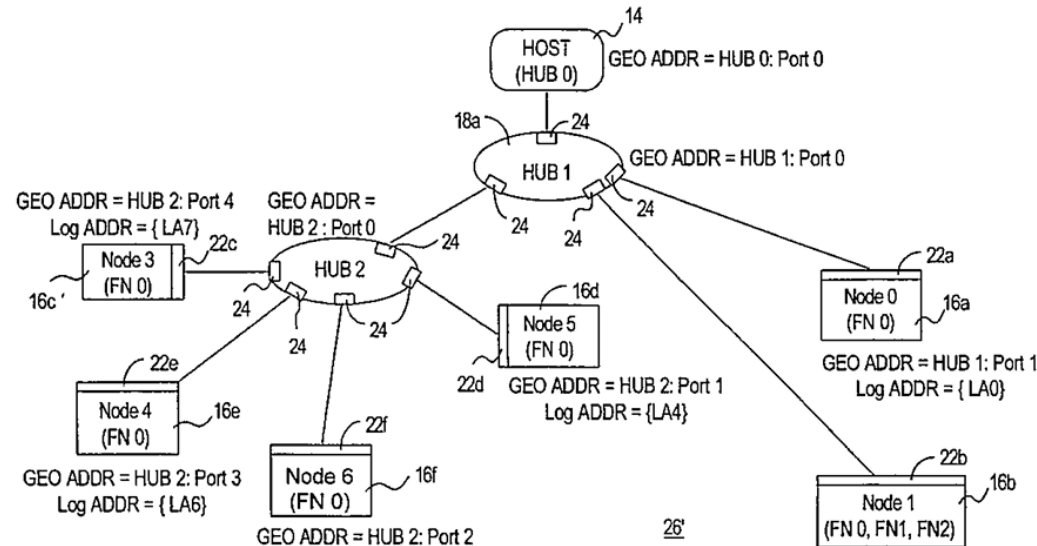
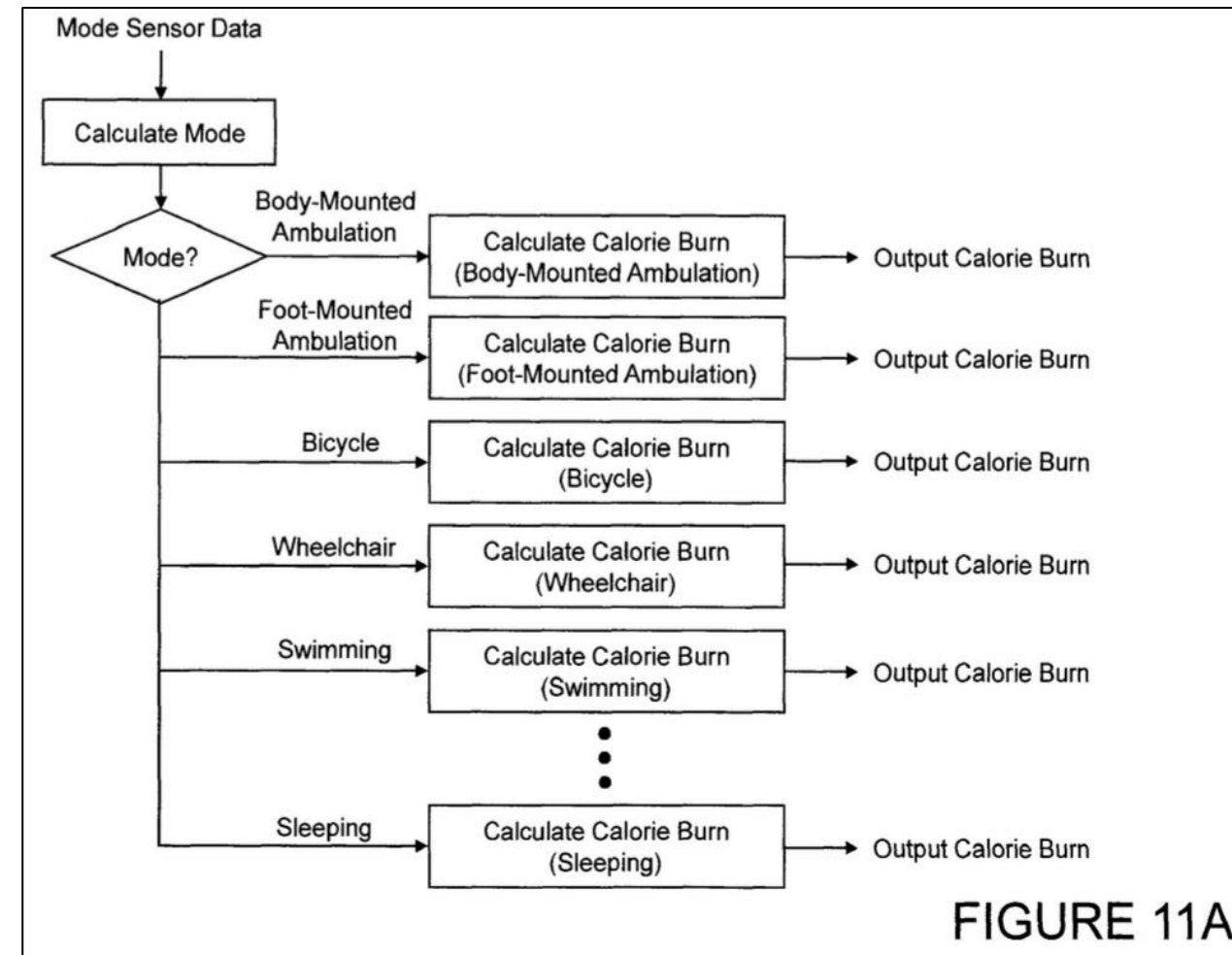
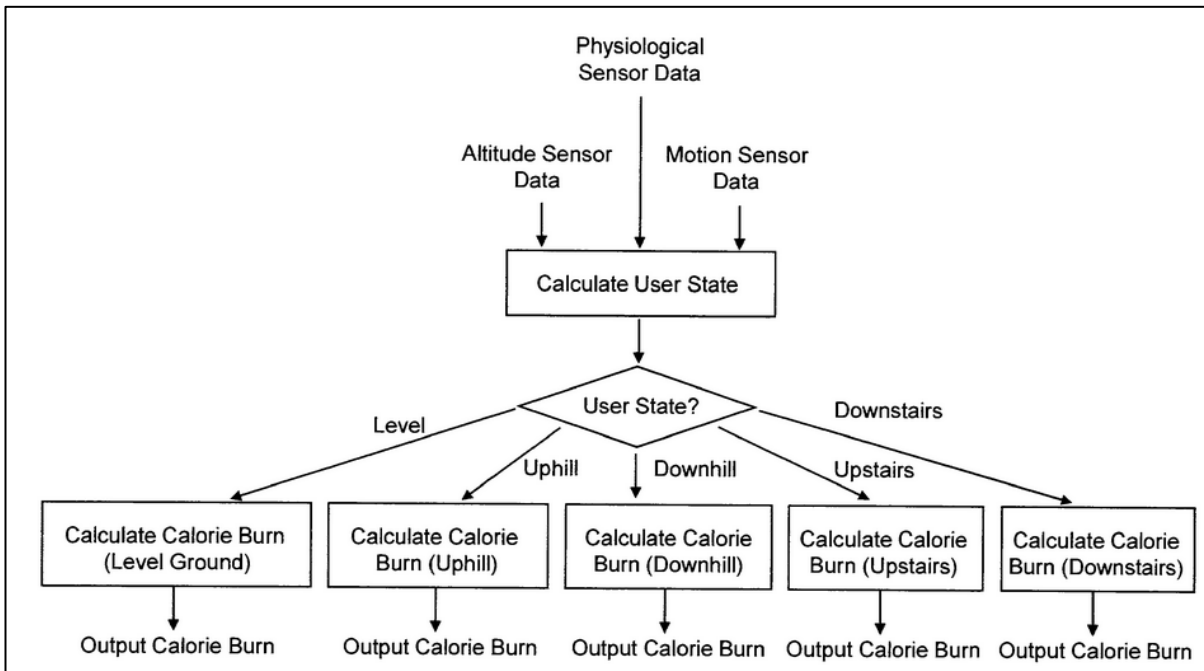
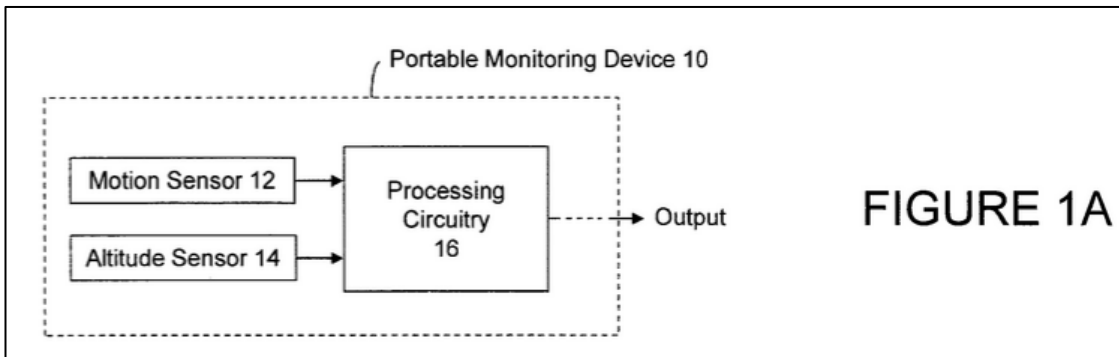


Figure 11

Example - Fitbit, Inc. US 8,180,591



Example - Fitbit, Inc. US 8,180,591

A portable monitoring device, ... comprising:

a housing having a physical size and shape . . . adapted to couple to the body of the user;

a motion sensor, disposed in the housing, ... wherein ... the motion sensor generates data which is representative of motion of the user;

an altitude sensor, disposed in the housing, ... wherein ... the altitude sensor generates data which is representative of a change in altitude of the user;

processing circuitry, disposed in the housing and coupled to the motion sensor and the altitude sensor, to:

generate the altitude sample signal in response to detecting a motion event using the data ... representative of motion of the user, and

calculate a number of stairs or flights of stairs traversed by the user using the data which is representative of a change in altitude of the user.



Patent Claims Example - Fitbit, Inc. US 8,180,591

FITBIT

CLAIM: A portable monitoring device, comprising:

a housing ...;

a motion sensor ...;

an altitude sensor ...;

processing circuitry... to:

generate the altitude sample signal ..., and
calculate a number of stairs or flights of stairs....

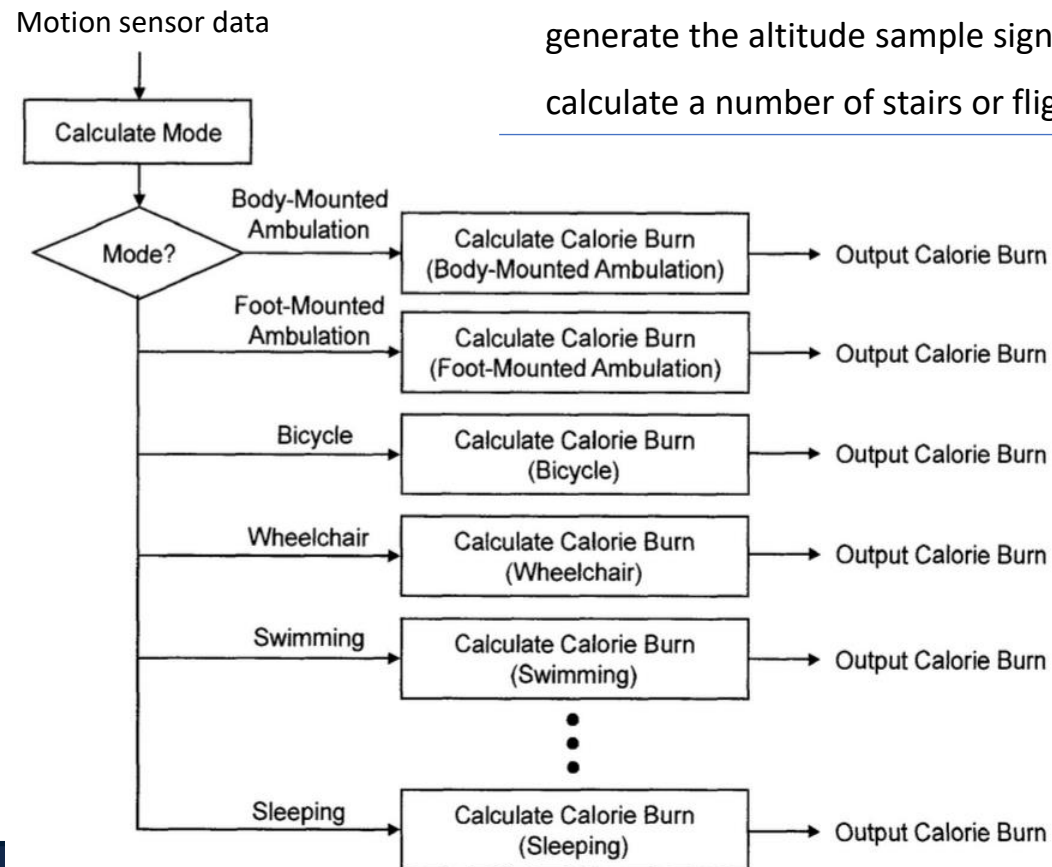


FIGURE 11A

OUR "INVENTION":

A health and fitness monitoring device:

- Includes:
 - a watch band;
 - a motion sensor in the band, and
 - electronics in the band
- Designed to:
 - identify a mode of movement of the user from the motion sensor signal,
 - determine an equation for calculating calorie burn based on the mode of movement, and
 - calculate a calorie burn of the user.

Is our device patentable?

No—the features are described in a previous publication

Example - Fitbit, Inc. US 8,180,591

CLAIM: A portable monitoring device, comprising:

a housing ...;

a motion sensor ...;

an altitude sensor ...;

processing circuitry... to:

generate the altitude sample signal ..., and

calculate a number of stairs or flights of stairs....

FITBIT

Motion sensor data

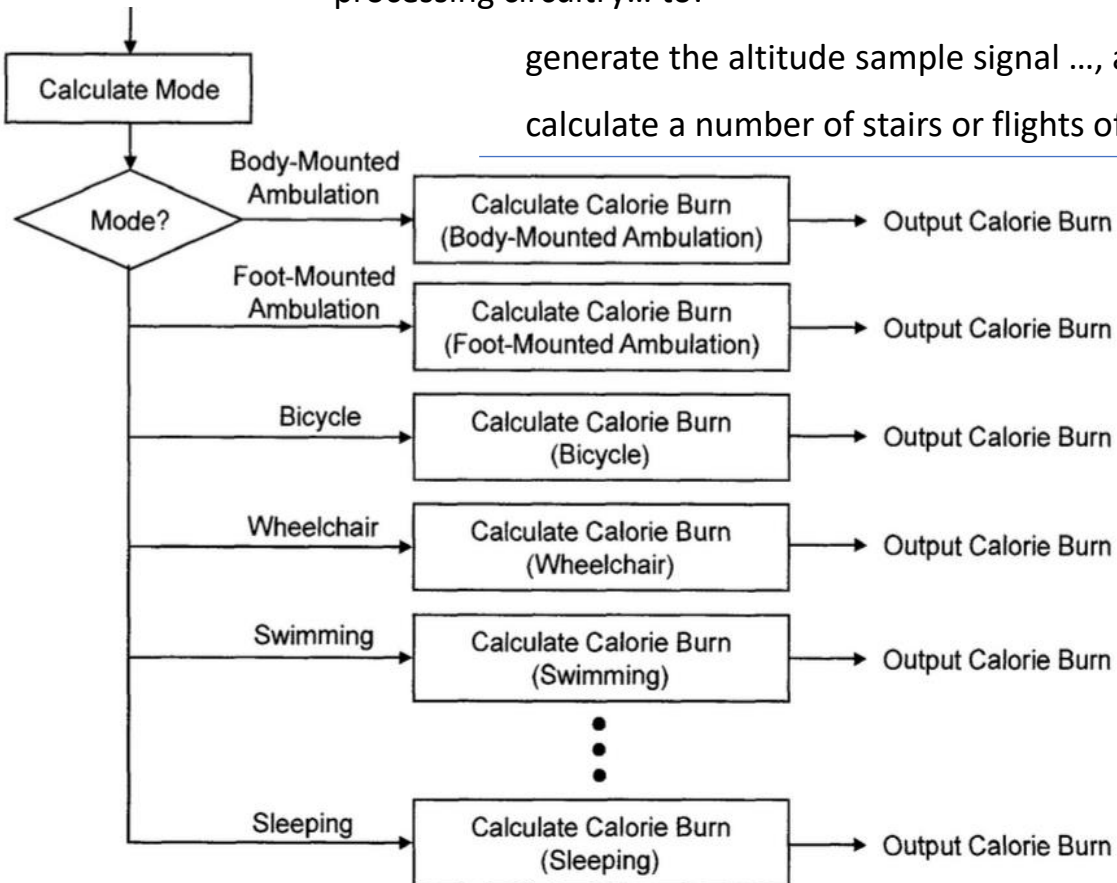


FIGURE 11A

OUR "INVENTION":

A health and fitness monitoring device:

- Includes:
 - a watch band;
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 - electronics in the band
- Designed to:
 - identify a mode of movement of the user from the motion sensor signal,
 - determine an equation for calculating calorie burn based on the mode of movement, and
 - calculate a calorie burn of the user.

Do we infringe?

No—our device is not covered by Fitbit's claims



MICHIGAN LAW
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Patent Claims Example - Fitbit, Inc. US 8,180,591

11. A portable monitoring device, comprising:

a housing ...;

a motion sensor ...;

an altitude sensor ...;

processing circuitry... to:

generate the altitude sample signal ..., and

calculate a number of stairs or flights of stairs....

FITBIT

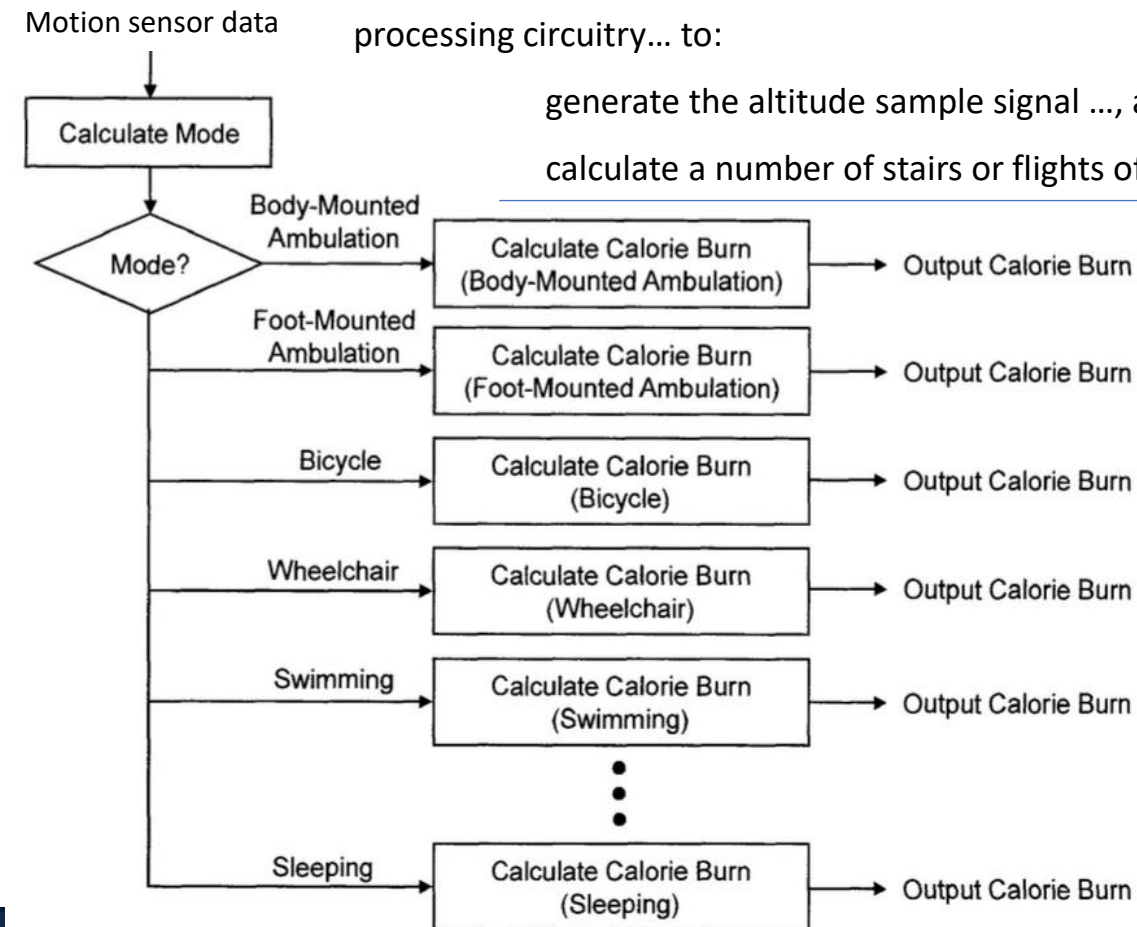


FIGURE 11A

OUR "INVENTION":

A watch:

- Includes:

- A motion sensor,
- An altitude sensor,
- **An audio output component, and**
- A computer with software that:
 - receives signals from the sensors,
 - automatically calculates the number of stairs climbed, and
 - automatically plays the Rocky theme song whenever 100 stairs are climbed.**

Do we infringe?

Yes—our device includes every element of Claim 11



MICHIGAN LAW
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Patent Claims Example - Fitbit, Inc. US 8,180,591

11. A portable monitoring device, comprising:

a housing ...;

a motion sensor ...;

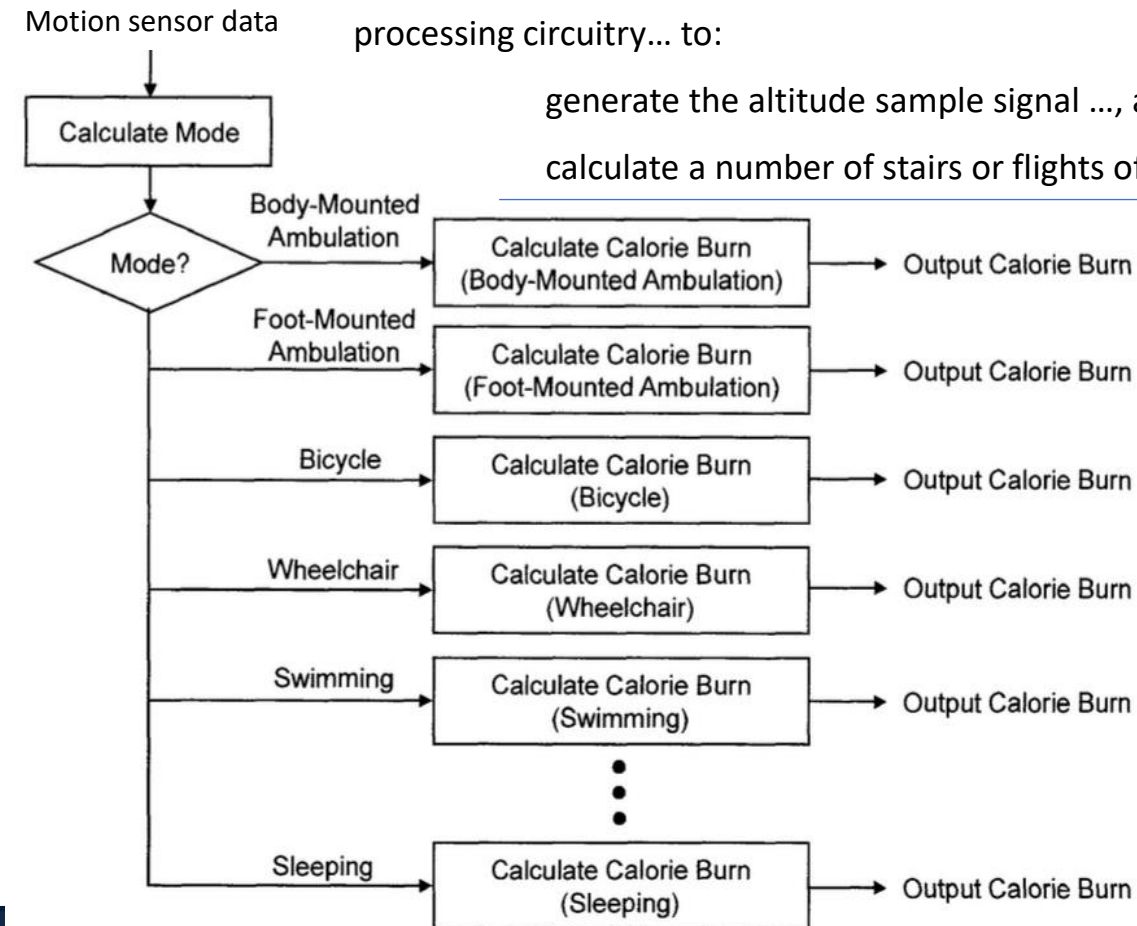
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processing circuitry... to:

generate the altitude sample signal ..., and

calculate a number of stairs or flights of stairs....

FITBIT



OUR "INVENTION":

A watch:

- Includes:

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- An altitude sensor,
- **An audio output component, and**
- A computer with software that:
 - receives signals from the sensors,
 - automatically calculates the number of stairs climbed, and
 - automatically plays the Rocky theme song whenever 100 stairs are climbed.**

Is this patentable?

Maybe; it has a unique feature not taught in the reference

FIGURE 11A

PATENT APPLICATION TIMING:

When to file is a balancing act

Need to:

- Be first to file
- **File before any public disclosure** (or within 1 year of disclosure if pursuing US protection only)



Need to:

- Have actual or constructive reduction to practice
- Be able to describe in sufficient detail to “enable”
- Have funds to pay for patent process

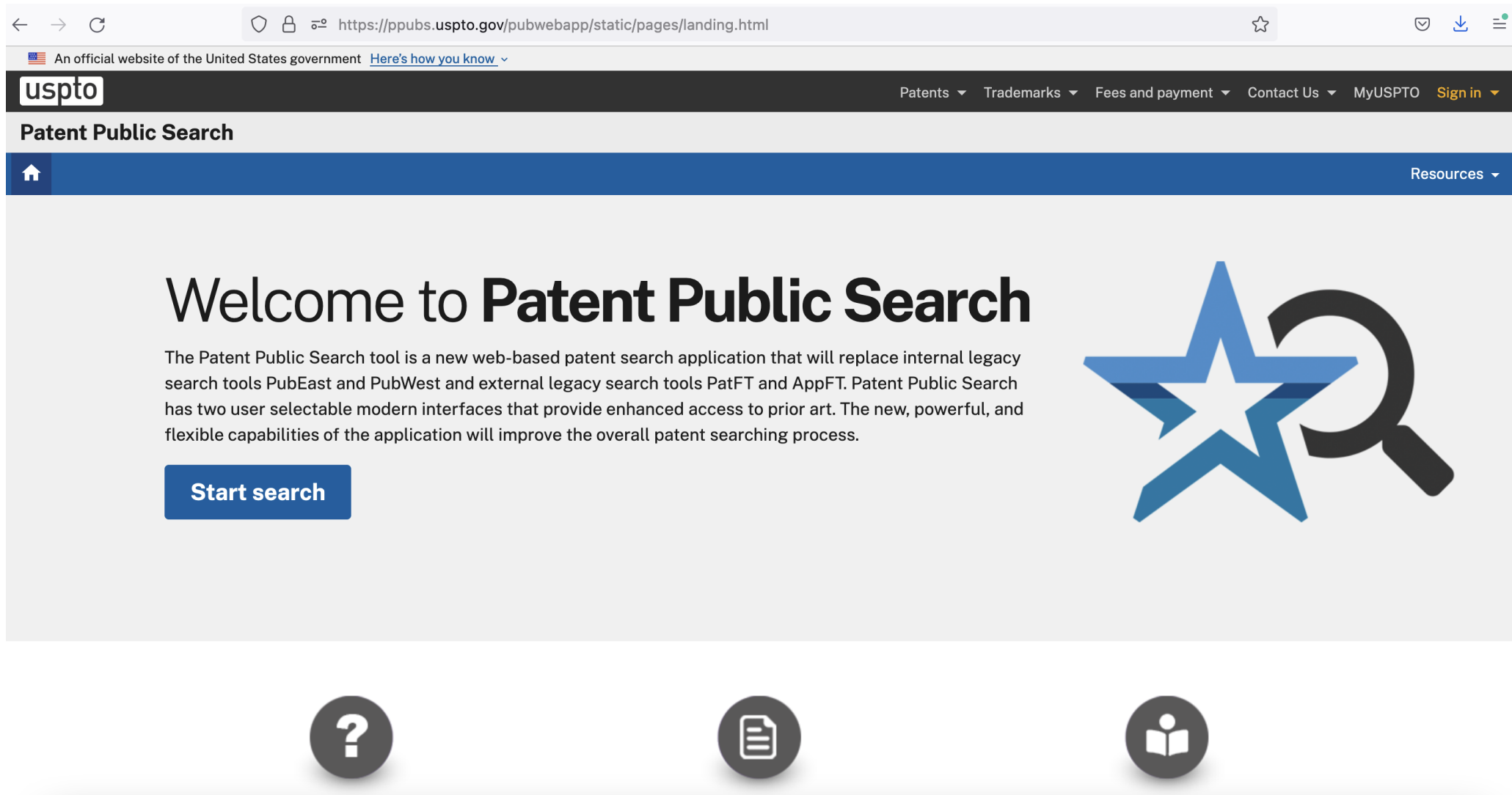


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About me

I support the teaching, research needs for the following areas of Engineering:

- Chemical Engineering
- Macromolecular Science and Engineering
- Materials Science and Engineering
- Patents and Trademarks
- Standards

I have been with the Engineering Library since January 1995. I have an M.L.S. degree from the Catholic University of America and an M.S. in Chemistry from the University of Mumbai. I am also the coordinator for collections for Art, Architecture, Engineering library. Previously, I worked at Gelman Sciences and American Tobacco Company.

Final Advice

Don't forget about the IP rights of others

- When developing a brand: Is your choice confusingly similar to an existing mark?
- When developing a website: Are there restrictions on the images you want to add?
- When developing software: Are you using open source code with restrictions on use?
- When developing a product: Does it infringe on an existing patent?
- When developing a side project: Are you obligated to assign the rights to your employer?



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THANK YOU!

Kristen Wolff
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